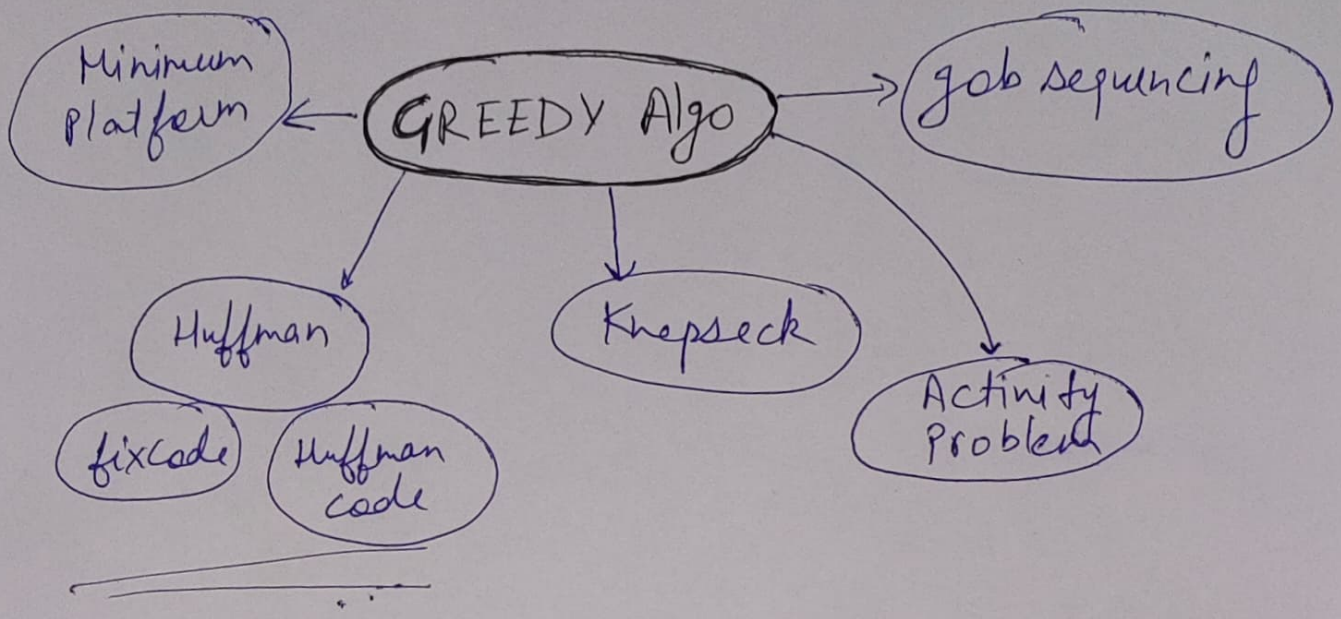
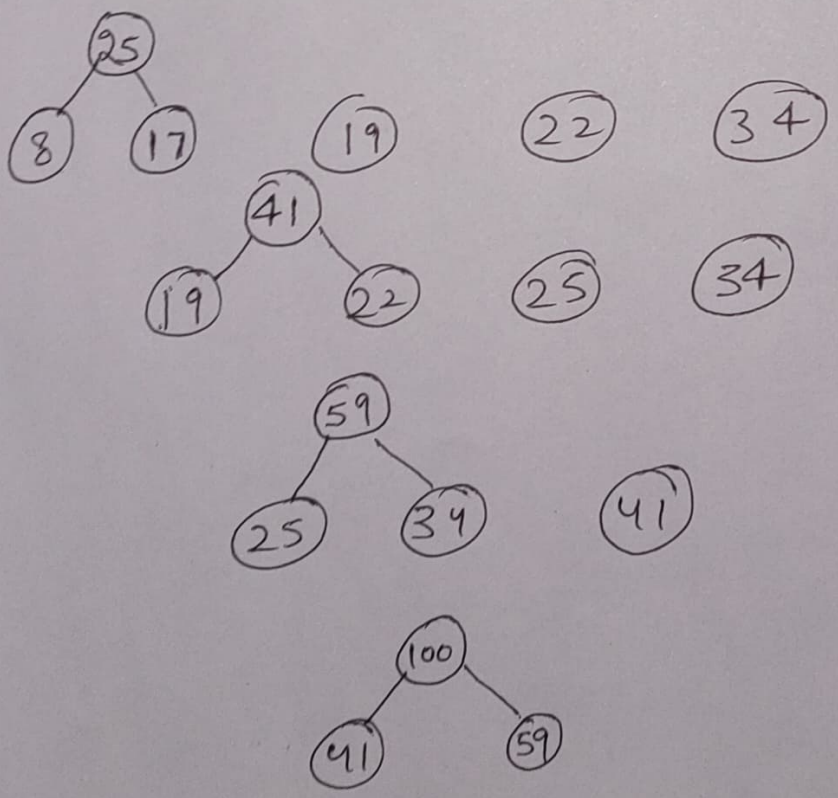

Minimum Platform
GREEDY Algo
Job sequencing
Huffman
Knapsack
Activity Problem
fixcode
Huffman code

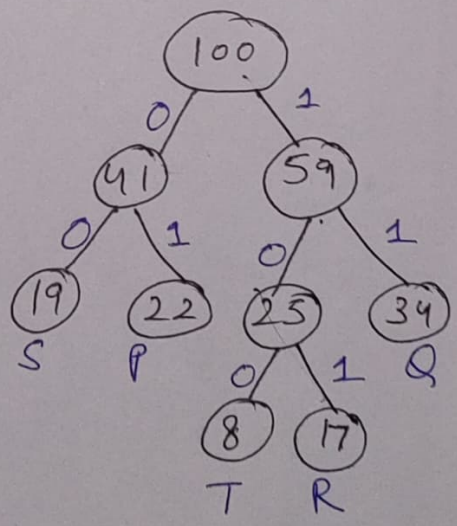
Q A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$

Character	Probability		
P	0.22	01	2×22
Q	0.34	11	2×34
R	0.17	101	3×17
S	0.19	00	2×19
T	0.08	100	3×8
	1.00		

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is

"RST" → [Huffman Code ↓ Decode.]





Q Consider a job scheduling problem with 4 jobs J_1, J_2, J_3, J_4 and with corresponding deadlines $(d_1, d_2, d_3, d_4) = (4, 2, 4, 2)$.
Which of the following is not a feasible schedule without violating any job schedule?

(a) J_2 J_4 J_1 J_3

(b) J_4 J_1 J_2 J_3

(c) J_4 J_2 J_1 J_3

(d) J_4 J_2 J_3 J_1

Q We are given 9 tasks $T_1, T_2 \dots T_9$. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has profit P_i and a deadline d_i Profit P_i is earned if the task is completed before the end of the d_ith unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

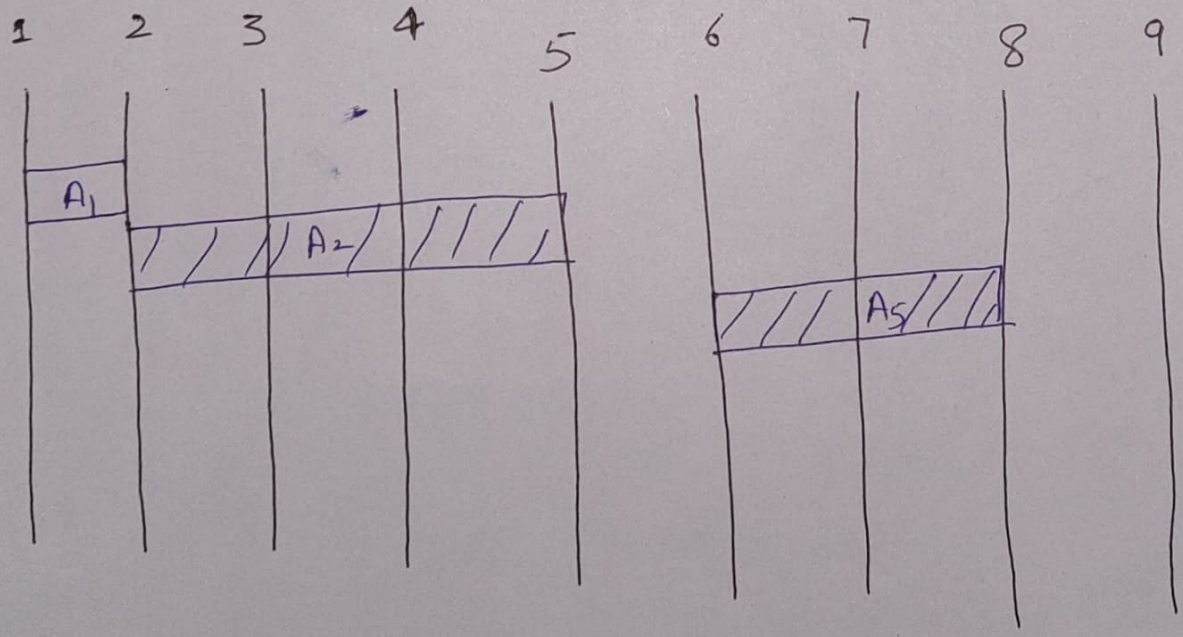
(8) (4) (1) (5) (6) 9 (3) (7) (2)

X

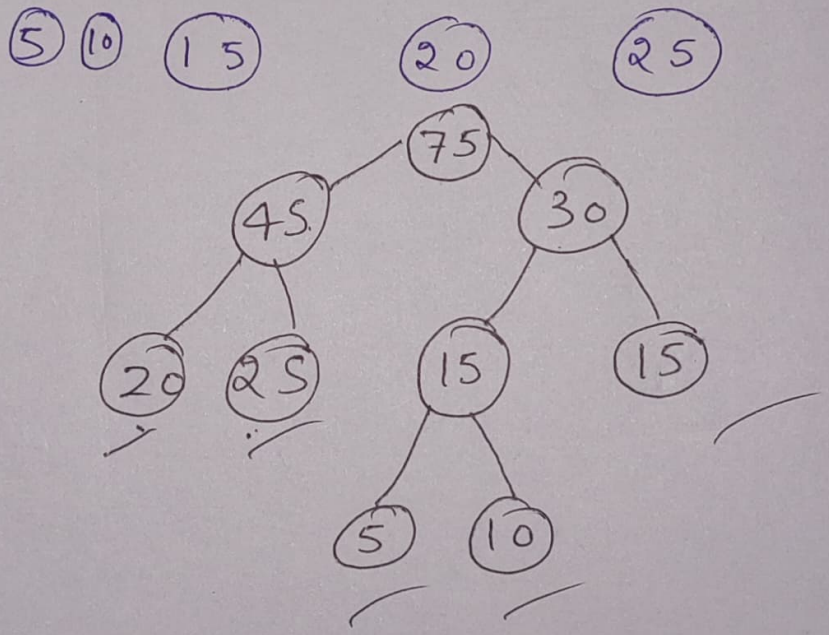
T_2	1
T_7	2
T_9	3
T_5	4
T_3	5
T_1	6
T_8	7

Q. Consider the following number of activities with their start and finish time given below. In which sequence will the activity be selected in order to maximize the number of activities without any conflicts?

Activity	starting time	finish time	
A_1	1	2	
A_2	2	5	
A_3	1	5	x
A_4	3	6	x
A_5	6	8	
A_6	4	9	



Q The minimum number of record movements required to merge five files A (with 10 records) B (20 records) C (15 records) D (5 records) E (25 records)



$= 20 \times 2 + 25 \times 2 + 5 \times 3 + 10 \times 3 + 15 \times 2$

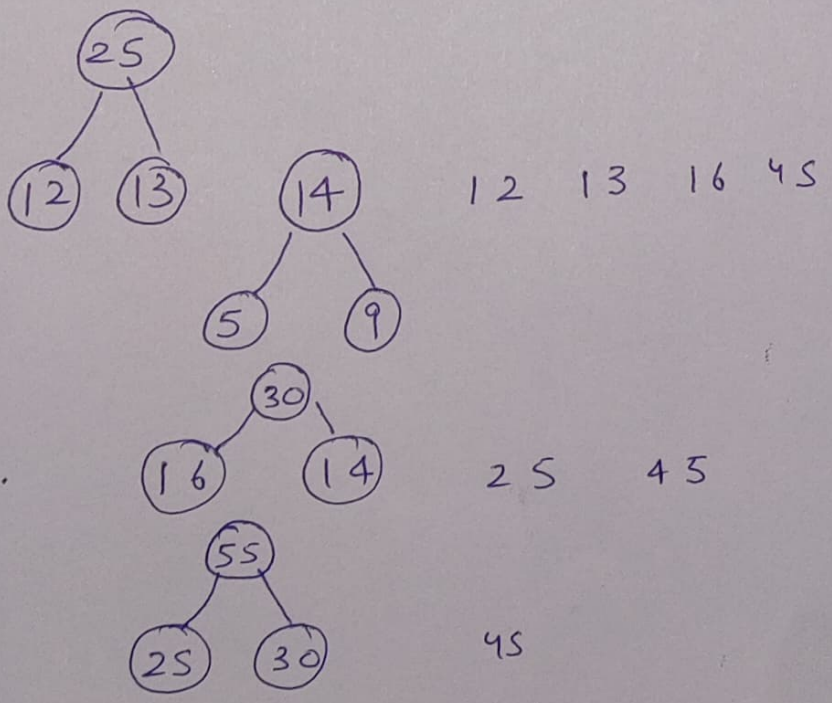
$= 165$

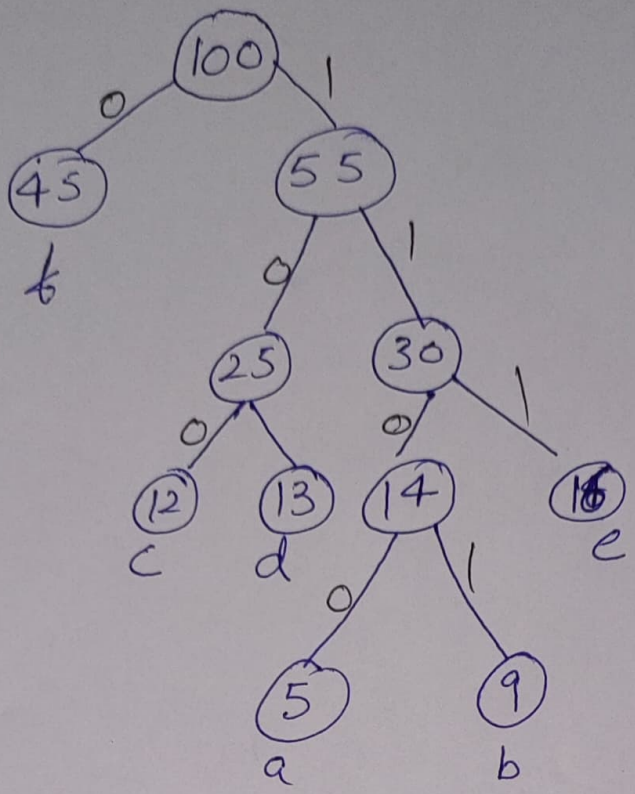
Q A networking company uses a compression technique to encode the message before transmitting over the network.

character	frequency
a	5
b	9
c	12
d	13
e	16
f	45

(a) If compression technique used is Huffman coding, how many bits will be saved in the message?

(b) What will be the Huffman encoding for "dead"




100
0
1
45
55
f
0
1
25
30
0
1
0
12
13
14
16
c
d
e
0
1
5
9
a
b